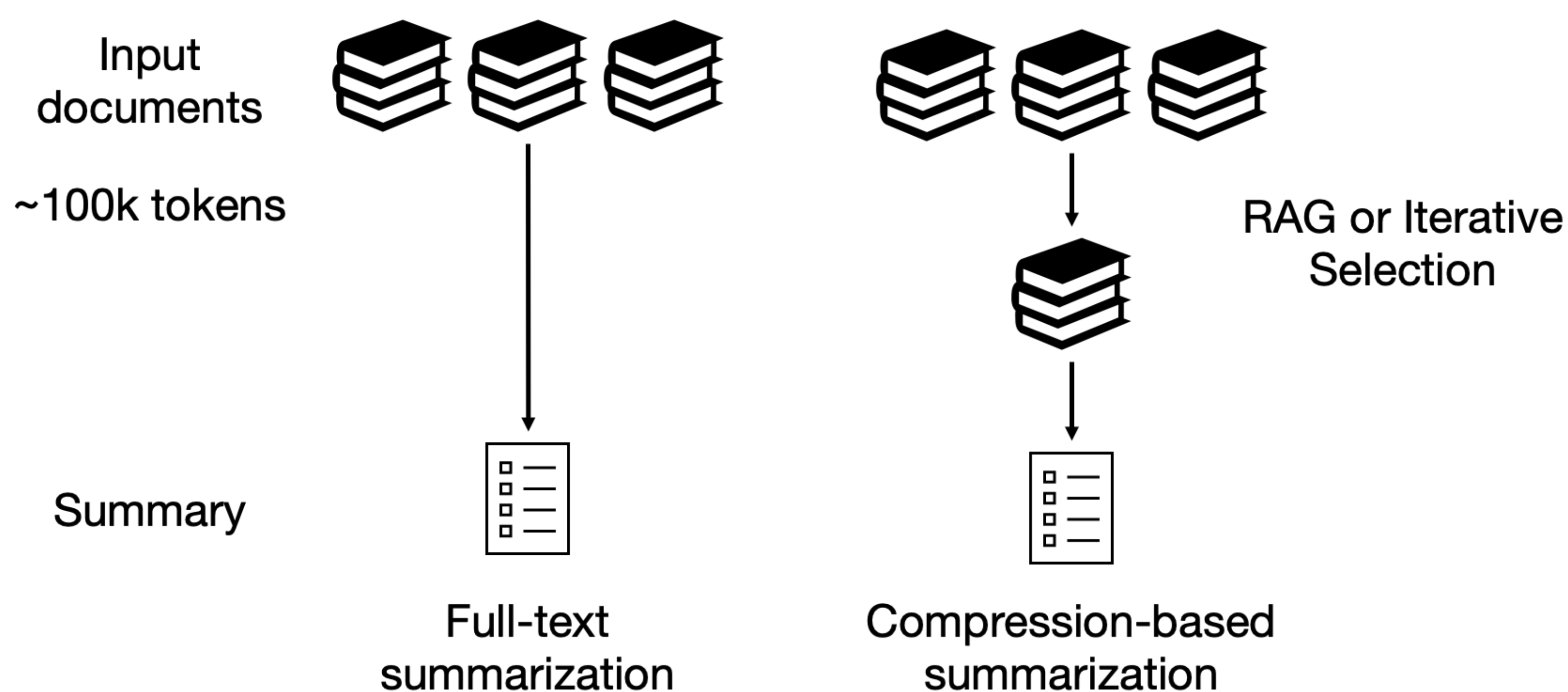


Scaling Multi-Document Event Summarization

Evaluating Compression vs. Full-Text Approaches



— Full input (w. long context window) or compressed input (w. short context window)? —



Compression

- Efficient
- Scalable
- Retrieval, Hierarchical, Incremental

Full-text

- Effective
- Utilizes long-context windows of recent LLMs



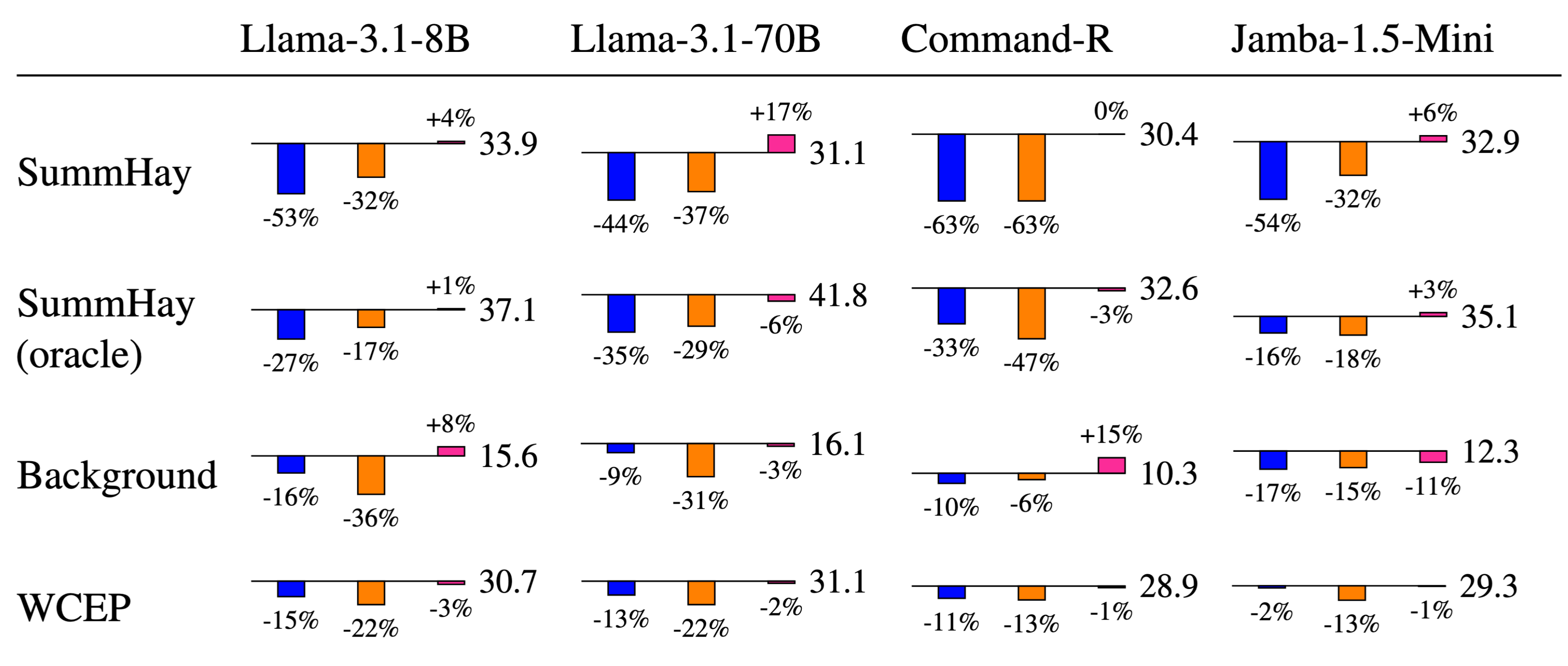
Base models

- Llama-3.1 (8B, 70B)
- Command-R (32B)
- Jamba 1.5 Mini (52B MoE, 12B active)

Datasets

- SummHay (Laban et al., 2024)
- Background (Pratapa et al., 2023)
- WCEP (Gholipour Ghalandari et al., 2020)

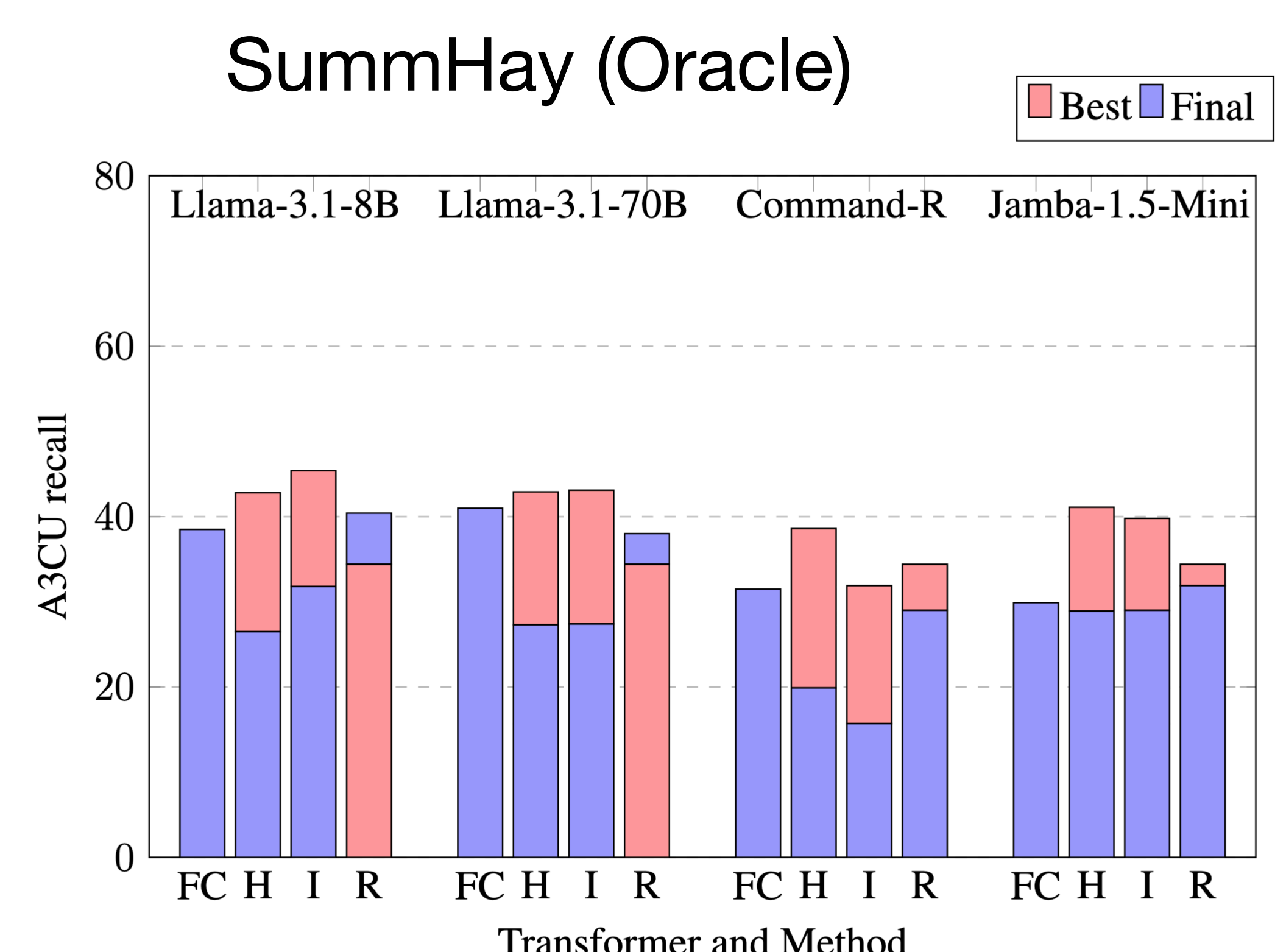
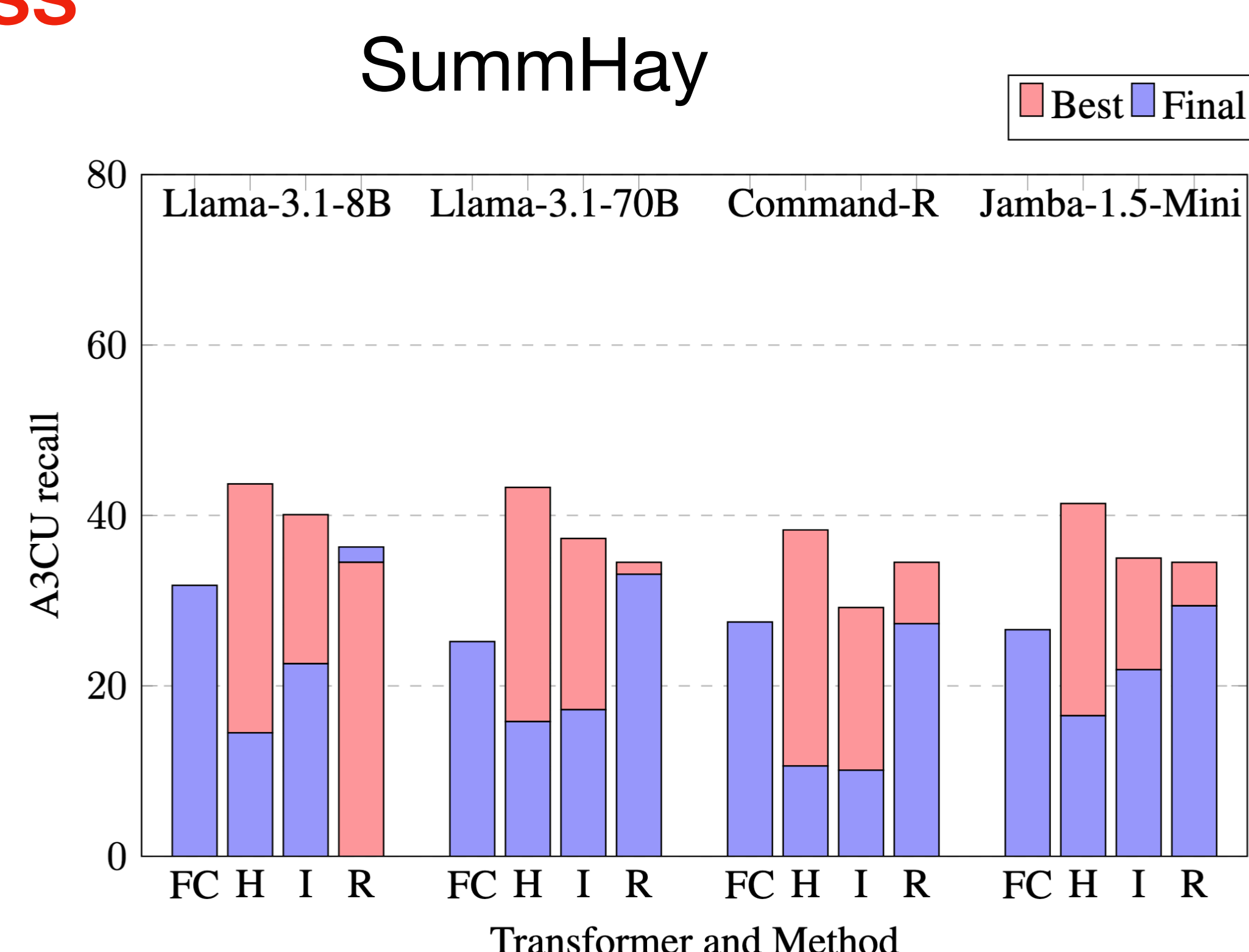
Results



Relative performances of **hierarchical**, **incremental** and **retrieval** compared to full-text baseline (AutoACU F1)

Tracking information loss

We measure A3CU recall scores of final system summary and the *best* intermediate output.



Proposals for future work

- Hybrid approaches that combine compression with long-context windows could be more efficient and effective.
- We need automatic metrics that do not rely on human-written references.

